上节回顾

静电场的标势　$\nabla\times\vec{E}=0 \Rightarrow \varphi(P)=\int_P^\infty \vec{E}\cdot d\vec{l}$　　$\vec{E}=-\nabla\varphi$

点电荷的电势：$\varphi(r)=\frac{q}{4\pi\varepsilon_0 r}$

连续电荷分布：$\varphi = \int_V \frac{\rho(x')\,dV'}{4\pi\varepsilon_0 r}$

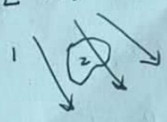
泊松方程：$\nabla^2\varphi=-\rho/\varepsilon$

边值条件　$\varphi_1=\varphi_2$

$$\varepsilon_2\frac{\partial\varphi_2}{\partial n}-\varepsilon_1\frac{\partial\varphi_1}{\partial n}=-\sigma_f$$

静电场的能量　$W=\frac{1}{2}\int\rho\varphi\,dV=\frac{\varepsilon_0}{2}\int_{全空间}\vec{E}^2\,dV$

相互作用能　$W=\int\varphi_1\rho_2\,dV$



静电场边值问题：

1) 分离变量法解拉普拉斯方程　$\nabla^2\varphi=0$

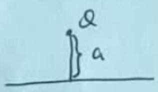
具有轴对称性的情况：

$$\varphi=\sum_n\left(a_nR^n+\frac{b_n}{R^{n+1}}\right)P_n(\cos\theta)$$

具有球对称　（取 $n=0$）

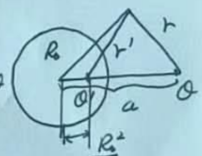
$$\varphi=a_0+\frac{b_0}{R}$$

2) 镜像法



(i) 点电荷在上半空间：$\varphi=\frac{1}{4\pi\varepsilon_0}\left[\frac{Q}{\sqrt{x^2+y^2+(z-a)^2}}-\frac{Q}{\sqrt{x^2+y^2+(z+a)^2}}\right]$

(ii) 导体球外有一点电荷　$\varphi=\frac{1}{4\pi\varepsilon_0}\left(\frac{Q}{r}+\frac{Q'}{r'}\right)$　　$Q'=-\frac{R_0}{a}Q$



3) 格林函数法

$$G|_S=0$$

第一类边值问题（给定边界的 φ）：$\varphi=\int_V G(x',x)\rho(x')\,dV'-\varepsilon_0\oint_S\left[\varphi(x')\frac{\partial G(x',x)}{\partial n'}\right]dS'$

第二…… …… $\frac{\partial\varphi}{\partial n}$：$\varphi=\int_V G(x',x)\rho(x')\,dV'+\varepsilon_0\oint_S G(x',x)\frac{\partial\varphi}{\partial n'}\,dS'+\langle\varphi\rangle_S$

$$\left.\frac{\partial G}{\partial n'}\right|_S=-\frac{1}{\varepsilon_0 S}$$

唯一性定理.

设区域V内给定自由电荷分布$\rho(x)$，在V的边界S上给定

1) 电势 $\varphi|_s$

或

2) 电势的法向偏导 $\frac{\partial\varphi}{\partial n}\Big|_s$

则V内的电场唯一地确定.

证明：设有两组不同的解φ'和φ''满足唯一性定理的条件. 令

$$\varphi = \varphi' - \varphi''$$

则由 $\nabla^2\varphi' = -\rho/\varepsilon_i$, $\nabla^2\varphi'' = -\rho/\varepsilon_i$, 得

$$\nabla^2\varphi = 0 \quad \text{（每个均匀区域 } V_i \text{ 内）}$$

在两均匀区域界面上有

$$\varphi_i = \varphi_j \quad \Rightarrow \varphi\text{是连续的}$$

$$\varepsilon_i\left(\frac{\partial\varphi}{\partial n}\right)_i = \varepsilon_j\left(\frac{\partial\varphi}{\partial n}\right)_j \quad \Rightarrow \varepsilon\frac{\partial\varphi}{\partial n}\text{是连续的}$$

在整个区域V的边界S上有

$$\varphi|_S = \varphi'|_S - \varphi''|_S = 0$$

或

$$\frac{\partial\varphi}{\partial n}\Big|_S = \frac{\partial\varphi'}{\partial n}\Big|_S - \frac{\partial\varphi''}{\partial n}\Big|_S = 0$$

$$0 = \oint_S \varepsilon(r)\varphi\nabla\varphi\cdot dS = \int_V \nabla\cdot(\varepsilon\varphi\nabla\varphi)\,dV$$

$$= \int_V \varepsilon(\nabla\varphi)^2 dV + \int_V \varepsilon\varphi\underbrace{\nabla^2\varphi}_{=0}dV$$

$$= \int_V \varepsilon(\nabla\varphi)^2 dV$$

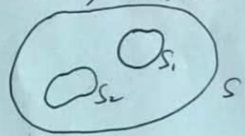
$$\nabla\varphi = 0$$

$$\varphi = \text{const}.$$

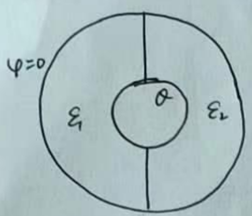
有导体存在时，需要给定导体上的电势，或导体上的总电荷 $\left(-\oint_{S_i}\frac{\partial\varphi}{\partial n}dS = \frac{Q_i}{\varepsilon}\right)$

证明过程与上类似. 只是

$$\oint_{S+S_i}\varepsilon(r)\varphi\nabla\varphi\cdot dS = 0 + \overset{\text{导体为等势体}}{\varphi_i}\oint_{S_i}\frac{\partial\varphi}{\partial n}dS = 0$$



例：两同心导体球壳之间充以两种介质，左半部电容率为ε_1，右半部电容率为ε_2，设内球壳带总电荷Q，外球壳接地，求电场和球壳上的电荷分布。



解：尝试解需满足边值关系

$$E_{2t}=E_{1t}$$
$$D_{2n}=D_{1n}$$

假设$\vec{E}$仍保持对称性

$$\vec{E_1}=\frac{A}{r^3}\vec{r}=\vec{E_1}=\vec{E}$$

满足上述边值条件

$$\oint_S \vec{D}\cdot d\vec{S}=\int_{S_1}\varepsilon_1\vec{E_1}\cdot d\vec{S}+\int_{S_2}\varepsilon_2\vec{E_2}\cdot d\vec{S}$$
$$=(\varepsilon_1+\varepsilon_2)2\pi\cancel{r^2}\cdot\frac{A}{\cancel{r^2}}=Q$$

$$A=\frac{Q}{2\pi(\varepsilon_1+\varepsilon_2)}$$

$$\therefore\quad \vec{E}=\frac{Q}{2\pi(\varepsilon_1+\varepsilon_2)r^3}\vec{r}$$

设内导体球半径为a，则球面上的自由电荷密度为

$$\sigma_1=D_{1r}=\varepsilon_1E_{1r}=\frac{\varepsilon_1Q}{2\pi(\varepsilon_1+\varepsilon_2)a^2}$$

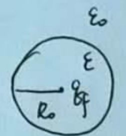
$$\sigma_2=D_{2r}=\varepsilon_2E_{2r}=\frac{\varepsilon_2Q}{2\pi(\varepsilon_1+\varepsilon_2)a^2}$$

自由电荷分布不同，为何$\vec{E}$保持球对称性？

$$\varepsilon_0E_{1r}=\sigma_1+\sigma_{1b}\Rightarrow\sigma_{1b}=(\varepsilon_0-\varepsilon_1)E_{1r}$$

$$\varepsilon_0E_{2r}=\sigma_2+\sigma_{2b}\Rightarrow\sigma_{2b}=(\varepsilon_0-\varepsilon_2)E_{2r}$$

例：均匀介质球的中心置一点电荷 Q_f，介质球的电容率为 ε，球外为真空，试用分离变量法求空间电势，把结果与使用高斯定理所得作对比。



解：点电荷的电场使介质球极化，总电势 φ 是点电荷产生的电势 φ_Q 与极化电荷产生的电势 φ' 之和，其中

$$\begin{cases} \nabla^2\varphi_1 = -Q_f\,\delta(x)/\varepsilon & (R<R_0) \\ \nabla^2\varphi_2 = 0 & (R>R_0) \\ R=0,\ \varphi_1\to\infty \\ R\to\infty,\ \varphi_2\to 0 \\ R=R_0,\ \varphi_1=\varphi_2 \\ \quad \varepsilon\dfrac{\partial\varphi_1}{\partial n} = \varepsilon_0\dfrac{\partial\varphi_2}{\partial n} \end{cases}$$

$$\varphi_Q = \begin{cases} \dfrac{Q_f}{4\pi\varepsilon R} & R<R_0 \\ \dfrac{Q_f}{4\pi\varepsilon_0 R} & R>R_0 \end{cases}$$

$$\varphi' = \begin{cases} a+\dfrac{b}{R} & R<R_0 \qquad R=0,\ \varphi'\text{有限} \Rightarrow b=0 \\ c+\dfrac{d}{R} & R>R_0 \qquad R\to\infty,\ \varphi'=0 \Rightarrow c=0 \end{cases}$$

$$= \begin{cases} a \\ \dfrac{d}{R} \end{cases}$$

边值关系：

$R=R_0$ 时 $\varphi_1=\varphi_2$，$\varepsilon\dfrac{\partial\varphi_1}{\partial R} = \varepsilon_0\dfrac{\partial\varphi_2}{\partial R}$

$$\begin{cases} \varphi_1 = \dfrac{Q_f}{4\pi\varepsilon R} + a \\ \varphi_2 = \dfrac{Q_f}{4\pi\varepsilon_0 R} + \dfrac{d}{R} \end{cases}$$

由 $R=R_0$ 时 $\varphi_1=\varphi_2$ 得

$$\frac{Q_f}{4\pi\varepsilon R_0} + a = \frac{Q_f}{4\pi\varepsilon_0 R_0} + \frac{d}{R_0} \quad \Rightarrow a = \frac{Q_f}{4\pi R_0}\left(\frac{1}{\varepsilon_0}-\frac{1}{\varepsilon}\right)$$

由 $\varepsilon\dfrac{\partial\varphi_1}{\partial R} = \varepsilon_0\dfrac{\partial\varphi_2}{\partial R}$ 得

$$-\frac{\varepsilon Q_f}{4\pi\varepsilon R_0^2} = -\frac{\varepsilon_0 Q_f}{4\pi\varepsilon_0 R_0^2} - \varepsilon_0\frac{d}{R_0^2} \quad \Rightarrow d=0$$

$$\therefore\ \varphi_1 = \frac{Q_f}{4\pi\varepsilon R} + \frac{(\varepsilon-\varepsilon_0)Q_f}{4\pi\varepsilon\varepsilon_0 R_0}$$

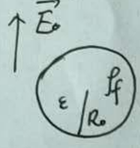
$$\varphi_2 = \frac{Q_f}{4\pi\varepsilon_0 R}$$

由高斯定理，球内外电场为

$$\vec{E}_1 = \frac{\vec{D}_1}{\varepsilon} = \frac{Q_f\vec{R}}{4\pi\varepsilon R^3} \qquad \vec{E}_2 = \frac{\vec{D}_2}{\varepsilon_0} = \frac{Q_f\vec{R}}{4\pi\varepsilon_0 R^3}$$

由电势为 $\varphi_1 = \int_R^{R_0}\vec{E}_1\cdot d\vec{R} + \int_{R_0}^{\infty}\cdot d\vec{R}$ $\qquad \varphi_2 = \int_R^{\infty}\vec{E}_2\cdot d\vec{R}$

例：在均匀外场E_0中置入一带均匀自由电荷密度为ρ_f的绝缘介质球（电容率为ε）
求空间各点的电势.



解：自由电荷的电场和外场E_0使介质球极化

$$\nabla^2\varphi_1 = -\rho_f/\varepsilon \qquad R<R_0$$

$$\nabla^2\varphi_2 = 0 \qquad R>R_0$$

$R=0$，φ_1有限

$R\to\infty$，$\varphi_2 \to -E_0R\cos\theta$

$R=R_0$，$\varphi_1=\varphi_2$，$\varepsilon\dfrac{\partial\varphi_1}{\partial R} = \varepsilon_0\dfrac{\partial\varphi_2}{\partial R}$

自由电荷的场和极化电荷的场都有球对称性.
由高斯定理.可求出球内外电场的球对称部分$E_{1\rho}$和$E_{2\rho}$
对球内：

$$\cancel{4\pi R^2}\,\varepsilon E_{1\rho} = \frac{4}{3}\cancel{\pi}R^3\rho_f$$

$$E_{1\rho} = \frac{R\rho_f}{3\varepsilon}$$

对球外

$$\cancel{4\pi}R^2\varepsilon_0E_{2\rho} = \frac{4}{3}\cancel{\pi}R_0^3\rho_f$$

$$E_{2\rho} = \frac{R_0^3\rho_f}{3R^2\varepsilon_0}$$

球外电势

$$\varphi_{2\rho} = \int_R^\infty E_{2\rho}\,dR = \frac{R_0^3\rho_f}{3\varepsilon_0}\left(-\frac{1}{R}\right)\Big|_R^\infty = \frac{\rho_f R_0^3}{3\varepsilon_0 R}$$

球内电势

$$\varphi_{1\rho} = \int_R^{R_0} E_{1\rho}\,dR + \int_{R_0}^\infty E_{2\rho}\,dR$$

$$= \frac{\rho_f}{3\varepsilon}\,\frac{1}{2}R^2\Big|_R^{R_0} + \frac{\rho_f R_0^3}{3\varepsilon_0 R_0}$$

$$= \frac{\rho_f}{6\varepsilon}(R_0^2-R^2) + \frac{\rho_f R_0^2}{3\varepsilon_0}$$

$\varphi_{1\rho}$是泊松方程$\nabla^2\varphi_1 = -\rho_f/\varepsilon$的特解.

$$\varphi_1 = \varphi_{1\rho}+\varphi_1', \qquad \varphi_2 = \varphi_{2\rho}+\varphi_2'$$

φ_1'和φ_2'为轴对称下拉普拉斯方程的通解.

$$\varphi_1' = \sum_n a_n R^n P_n(\cos\theta) \qquad R<R_0$$

$$\varphi_2' = \sum_n \frac{b_n}{R^{n+1}} P_n(\cos\theta) + \varphi_0 \qquad R>R_0$$

由 $R\to\infty$ $\varphi_2 = -E_0 R\cos\theta$ 得 $\varphi_0 = -E_0 R\cos\theta$

由 $R=R_0$ 时，$\varphi_1=\varphi_2$ 得

$$\cancel{\frac{\rho_f R_0^2}{3\varepsilon_0}} + \sum_n a_n R_0^n P_n(\cos\theta) = \cancel{\frac{\rho_f R_0^2}{3\varepsilon_0}} - E_0 R_0 \cos\theta + \sum_n \frac{b_n}{R_0^{n+1}} P_n(\cos\theta)$$

比较 P_1 的系数得

$$a_1 R_0 = -E_0 R_0 + \frac{b_1}{R_0^2} \quad \cdots\cdots (1)$$

由 $R=R_0$ 时 $\varepsilon \frac{\partial \varphi_2}{\partial R} = \varepsilon_0 \frac{\partial \varphi_1}{\partial R}$ 得

$$\varepsilon \sum_n a_n n R_0^{n-1} P_n(\cos\theta) = -\varepsilon_0 E_0 \cos\theta - \varepsilon_0 \sum_n (n+1)\frac{b_n}{R_0^{n+2}} P_n(\cos\theta)$$

比较 P_1 的系数得

$$\varepsilon a_1 = -\varepsilon_0 E_0 - \varepsilon_0 \, 2\frac{b_1}{R_0^3} \quad \cdots\cdots (2)$$

联合(1).(2)可得

$$a_1 = -\frac{3\varepsilon_0}{\varepsilon+2\varepsilon_0}E_0, \quad b_1 = R_0^3 E_0 \frac{\varepsilon-\varepsilon_0}{\varepsilon+2\varepsilon_0}$$

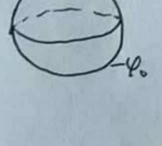
因此:

$$\varphi_1 = \frac{\rho_f (R_0^2-R^2)}{6\varepsilon} + \frac{\rho_f R_0^2}{3\varepsilon_0} - \frac{3\varepsilon_0}{\varepsilon+2\varepsilon_0}E_0 R\cos\theta$$

$$\varphi_2 = \frac{\rho_f R_0^3}{3\varepsilon_0 R} - E_0 R\cos\theta + \frac{(\varepsilon-\varepsilon_0)E_0 R_0^3}{(\varepsilon+2\varepsilon_0)R^2}\cos\theta$$

例：一半径为 R_0 的球面，在球坐标 $0<\theta<\frac{\pi}{2}$ 的半球面上电势为 φ_0，在 $\pi/2<\theta<\pi$ 的半球面上电势为 $-\varphi_0$，求用格林函数法求空间各点的电势.

第一类边值问题.



$$\varphi(x) = \underbrace{\int_V G(x',x)\rho(x')\,dV'}_{0} - \varepsilon_0 \oint_S \left[\varphi(x')\frac{\partial G(x',x)}{\partial n'}\right] dS'$$

其中 $G(x',x)$ 为球面电势为0时 x'处 单位点电荷在 x处 产生的电势